

Case Study Analysis: Targeting Prime Vulnerability Moments

Societal/Environmental Impact & Sustainability

The system proposed in the case study, which uses AI to identify and target individuals during their "prime vulnerability moments," has a profoundly negative net effect on society and raises serious questions about its sustainability.

Impact on People Involved Directly:

Directly, the system affects consumers by exploiting their psychological states for commercial gain. As the case mentions, a person feeling less attractive or fatigued may be deliberately targeted with ads for beauty or wellness products. This preys on negative emotions, potentially exacerbating feelings of insecurity and creating a cycle where individuals seek emotional solace in consumption. This is not empowerment but a form of psychological manipulation that can lead to financial strain and buyer's remorse, undermining genuine well-being (Rosen, 2013).

Long-Term Effects:

In the long term, the normalization of such practices erodes consumer autonomy and trust. If individuals become aware that their digital interactions are constantly being mined for moments of weakness, it could lead to widespread digital distrust and anxiety. Societally, it reinforces material consumption as a solution to emotional problems, which is environmentally unsustainable.

Sustainability:

This system is not sustainable. Psychologically, it is damaging. Environmentally, it promotes overconsumption. Socially, it is corrosive to trust. A sustainable business model should create value for all stakeholders, including consumers, without resorting to exploitative practices.

Since Zuboff (2019) claims in her book on surveillance capitalism, the logic behind such systems is that they should be highly efficient in harvesting human experience so as to be monetized, which, again, is logically incompatible with long-term social and individual well-being.

Ethical Violations: Bias, Transparency, and Accountability

The system has significant ethical violations, particularly concerning bias and a lack of transparency and accountability.

Bias:

The system is inherently biased against any group that can be algorithmically identified as being in a vulnerable state. The original study focused on US women, but an AI-driven system could expand this to target any demographic individuals grieving a loss, those with mental health challenges, or people experiencing financial distress.

Transparency:

The system is fundamentally non-transparent. The core of its business model relies on the consumer not knowing how their data is being used to manipulate them. As Mittelstadt et al. (2016) note in their seminal paper, the "opacity" of AI systems—the inability for a subject to understand how a decision about them was made is a key ethical challenge. The company would never disclose it is targeting "vulnerable moments" as it would rightly cause public outrage and render the tactic ineffective.

Accountability:

The company avoids accountability by using the positive framing mentioned in the case study terms like "encourage" and "empower." This allows them to deflect criticism and claim they are providing a beneficial service. There is no mechanism for a user to question why they were shown a specific ad at a specific time, and the company would likely deny any manipulative intent, holding itself unaccountable for the potential harm caused.

Legal Implications and Engineer's Responsibilities

The deployment of this system carries serious legal implications and places heavy responsibilities on the engineers who build it.

Legal Implications and Repercussions:

If deployed, particularly in regions with strong data protection laws like the European Union, this system would likely violate several legal principles. The EU's General Data Protection Regulation (GDPR) mandates that processing of personal data must be fair and lawful. Exploiting a data subject's vulnerability for gain would almost certainly be considered unfair processing. It could also violate rules against using special categories of data (e.g., data revealing psychological state) without explicit consent. Legal repercussions could include massive fines (up to 4% of global turnover under GDPR), injunctions stopping the practice, and class-action lawsuits from consumers claiming manipulation and harm.

Responsibilities of an Engineer:

An engineer has a professional and ethical duty to ensure safety, fairness, and compliance before deploying a system. In this case, they must be aware of:

Safety: The psychological safety of the end-user. Could this system exacerbate anxiety, depression, or financial problems?

Fairness: Is the system designed to exploit rather than serve? Engineers must implement ethical risk assessments and fairness audits.

Compliance: They must be intimately familiar with relevant regulatory frameworks like GDPR, the Digital Services Act (DSA), or the California Consumer Privacy Act (CCPA), which aim to protect users from such manipulative practices.

The engineer's responsibility extends beyond mere code functionality. They should advocate for ethical design principles, question the project's goals, and potentially refuse to work on a system they deem exploitative, as outlined in professional codes of ethics. Wachter & Mittelstadt (2021) discuss the need for a "right to reasonable inferences," meaning systems should not be allowed to make and act upon manipulative inferences about users, a concept an engineer must be aware of.

Reference:

1. Zuboff, Shoshana. *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. New York: PublicAffairs, 2019.
2. Mittelstadt, B. D., Allo, P., Taddeo, M., Wachter, S., & Floridi, L. (2016). The ethics of algorithms: Mapping the debate. *Big Data & Society*, 3(2). <https://doi.org/10.1177/2053951716679679>
3. Wachter, Sandra and Mittelstadt, Brent, A Right to Reasonable Inferences: Re-Thinking Data Protection Law in the Age of Big Data and AI (October 5, 2018). *Columbia Business Law Review*, 2019(2), Available at SSRN: <https://ssrn.com/abstract=3248829>